$A(W^T \cdot x_i)$

$3*0.5 + 3*0.6$

$+ 2*0.7$

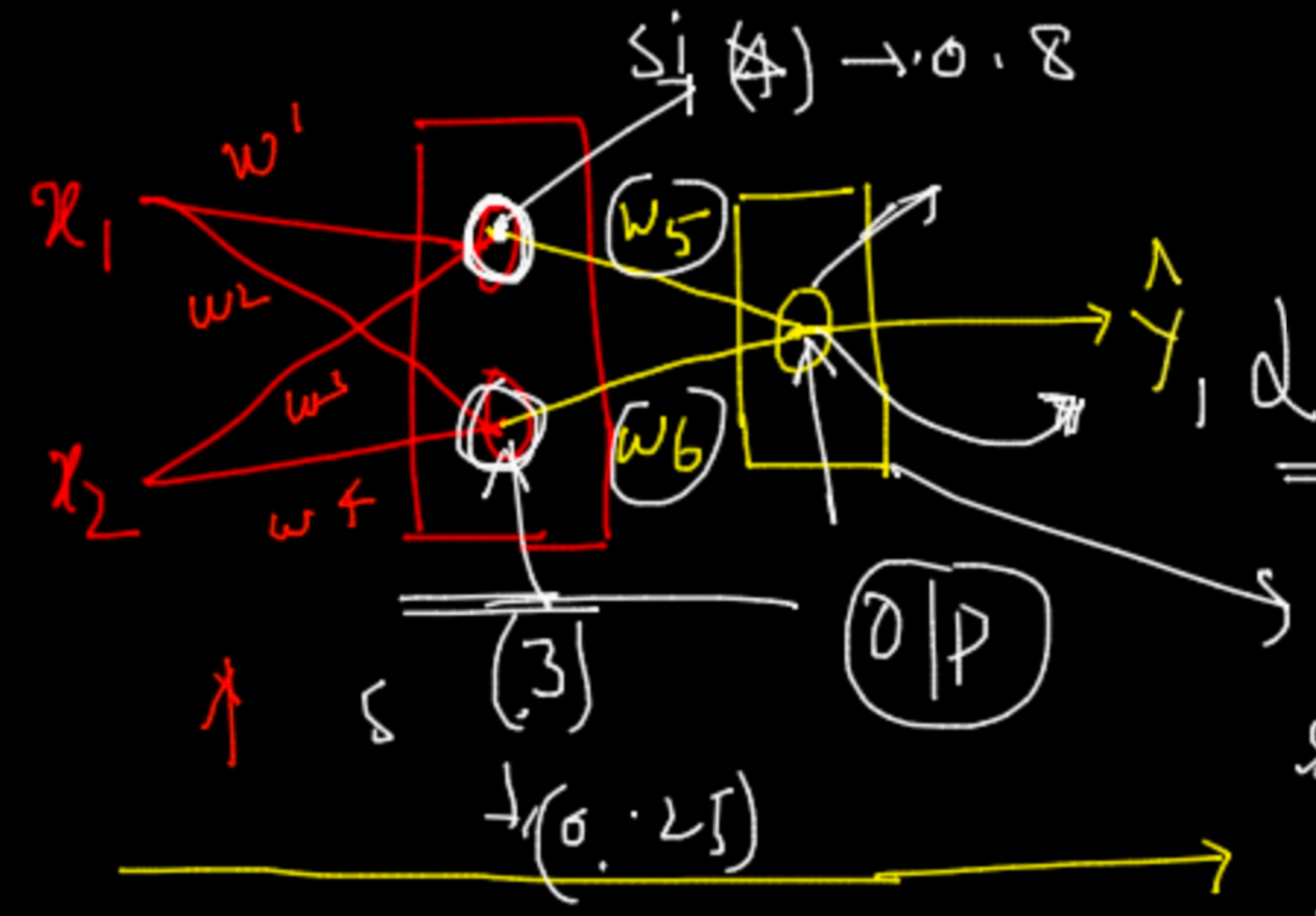
$+ 2*0.8$

$sig(4) \Rightarrow (0-1) \Rightarrow (0.8)$

$(0-1)$

if 0.57 →

x_1	x_2	~~y~~	$\hat{y}$
3	2	155	170
4	5	160	165
6	7	=	170



$\hat{y}, d(y, \hat{y})$

$sig(0.7) \Rightarrow 0.7$ class 1

70%

| Σ | f |

30% → 0

$W_1\ W_2\ W_3\ W_4$

$[0.5\ 0.6\ 0.7\ 0.8]$

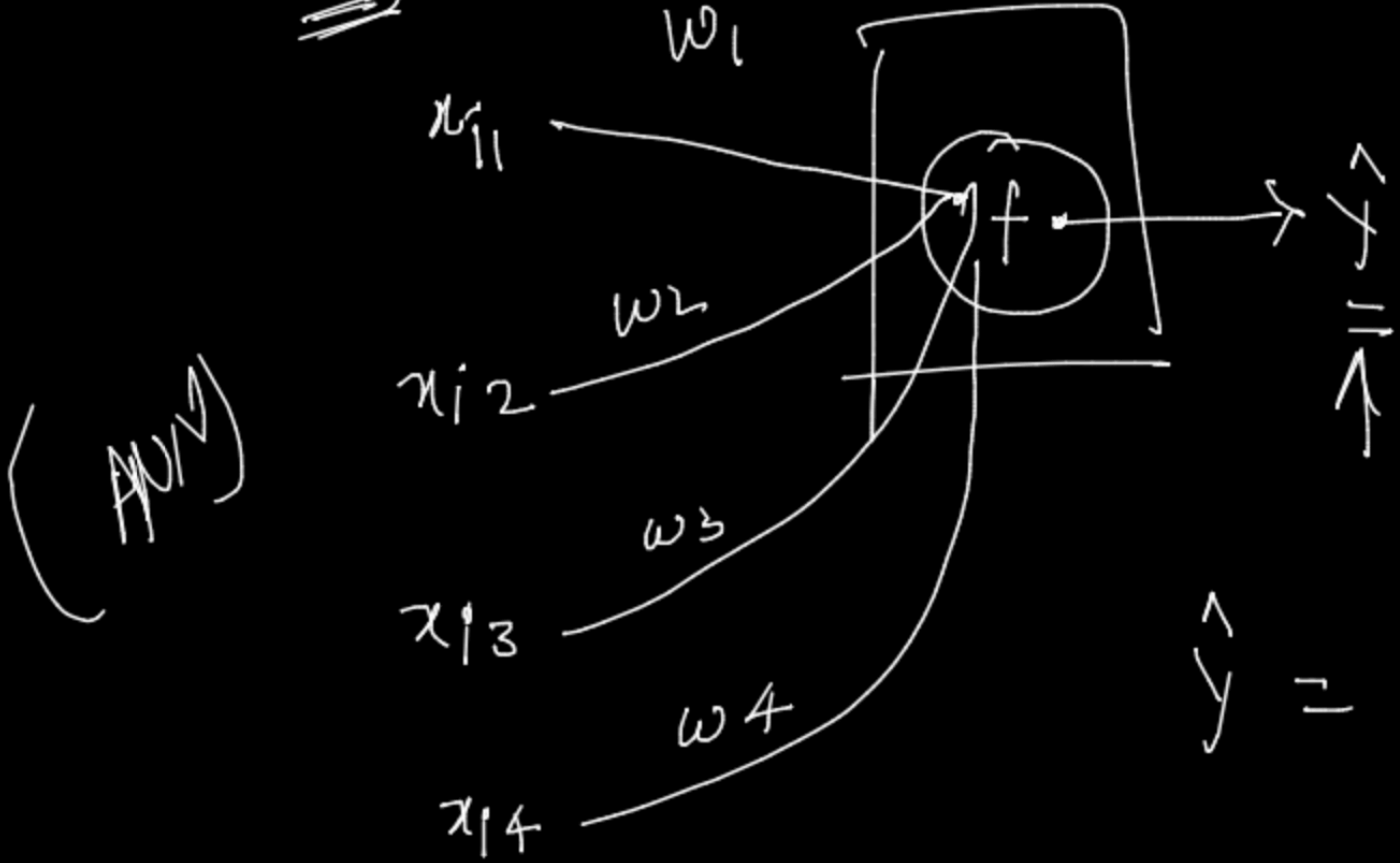
$W_5\ W_6$

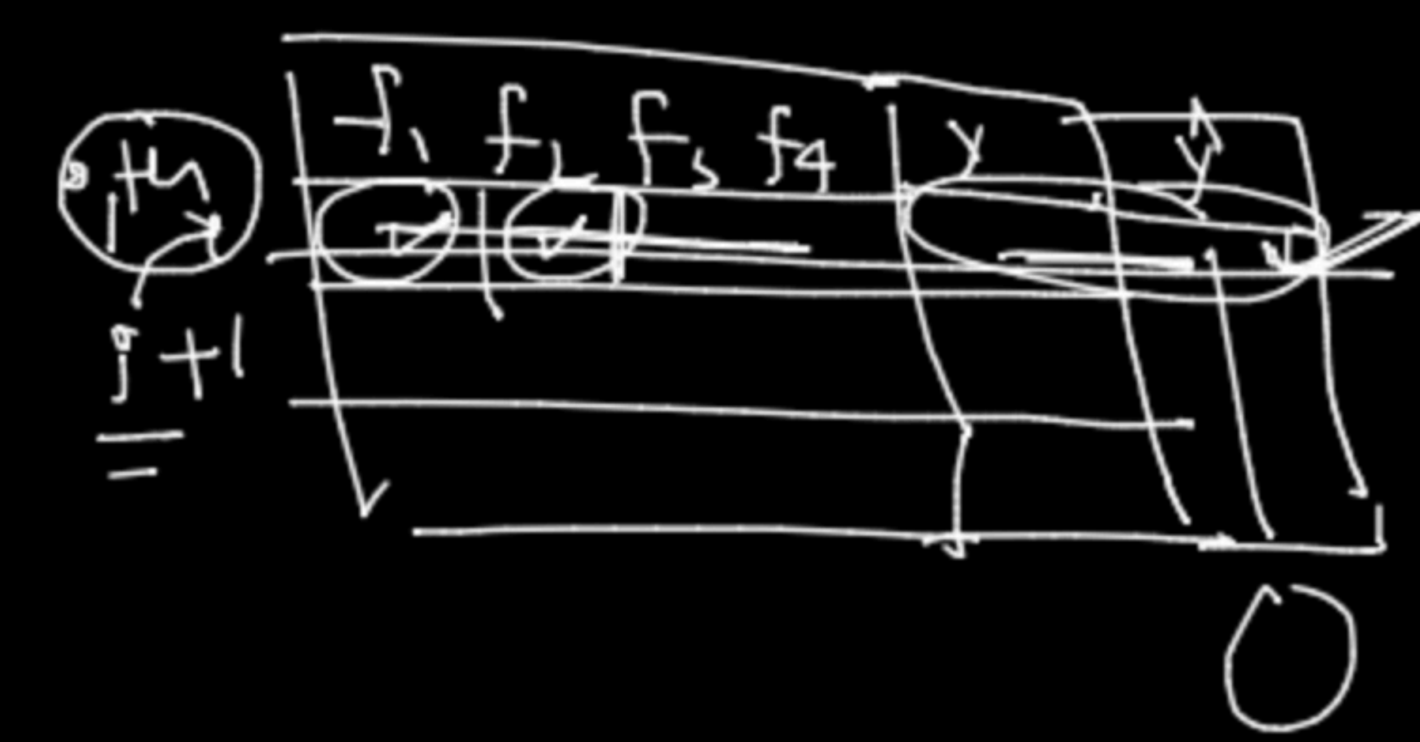
$[0.7, 0.8]$

$f(W^T \cdot x_i)$

sign

input = 4, hidden layers, Neuron = 1





Code

x_{i1}

$$\hat{y} = w^T \cdot x_i + \lambda w^T \cdot w$$

(L2 / L1)

Linear Regression

$$\min_{w^*} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 + reg$$

(Step1) Define loss function (eg - loss) (linear Regression) w*

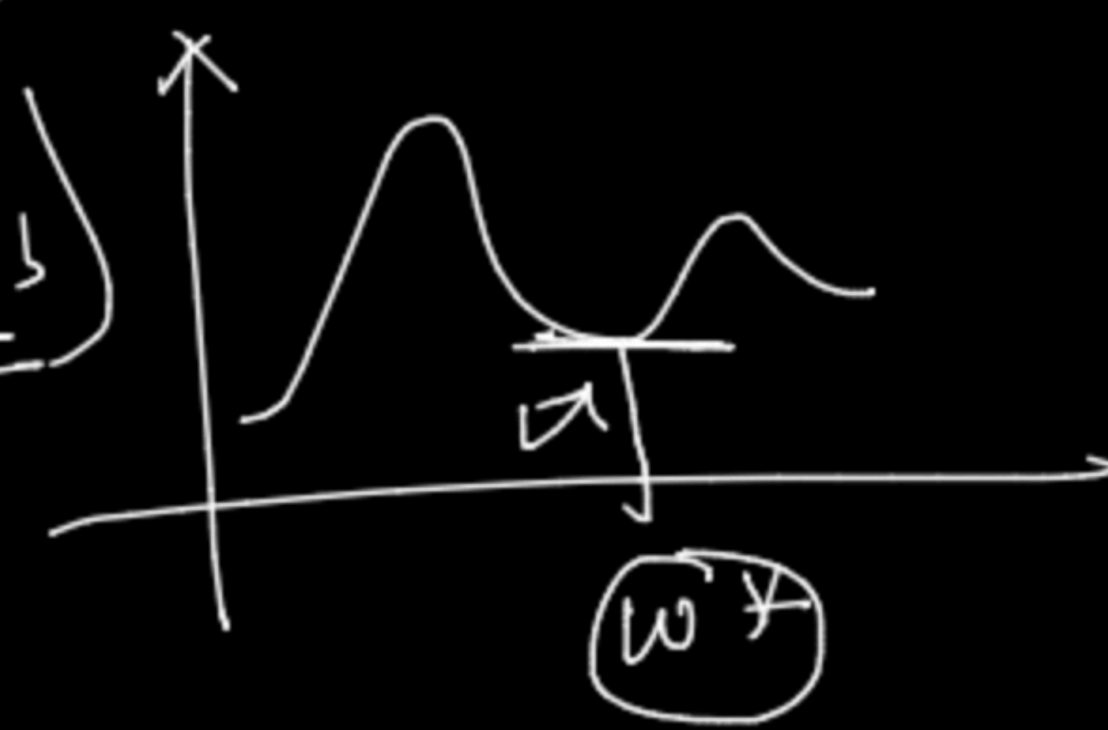
Cost fn

$$d = \sum_{i=1}^{n} (y_i - \hat{y})^2 + reg$$

class (cross-Entropy)

— $\hat{y}$ y

f(loss)



maxima & minima

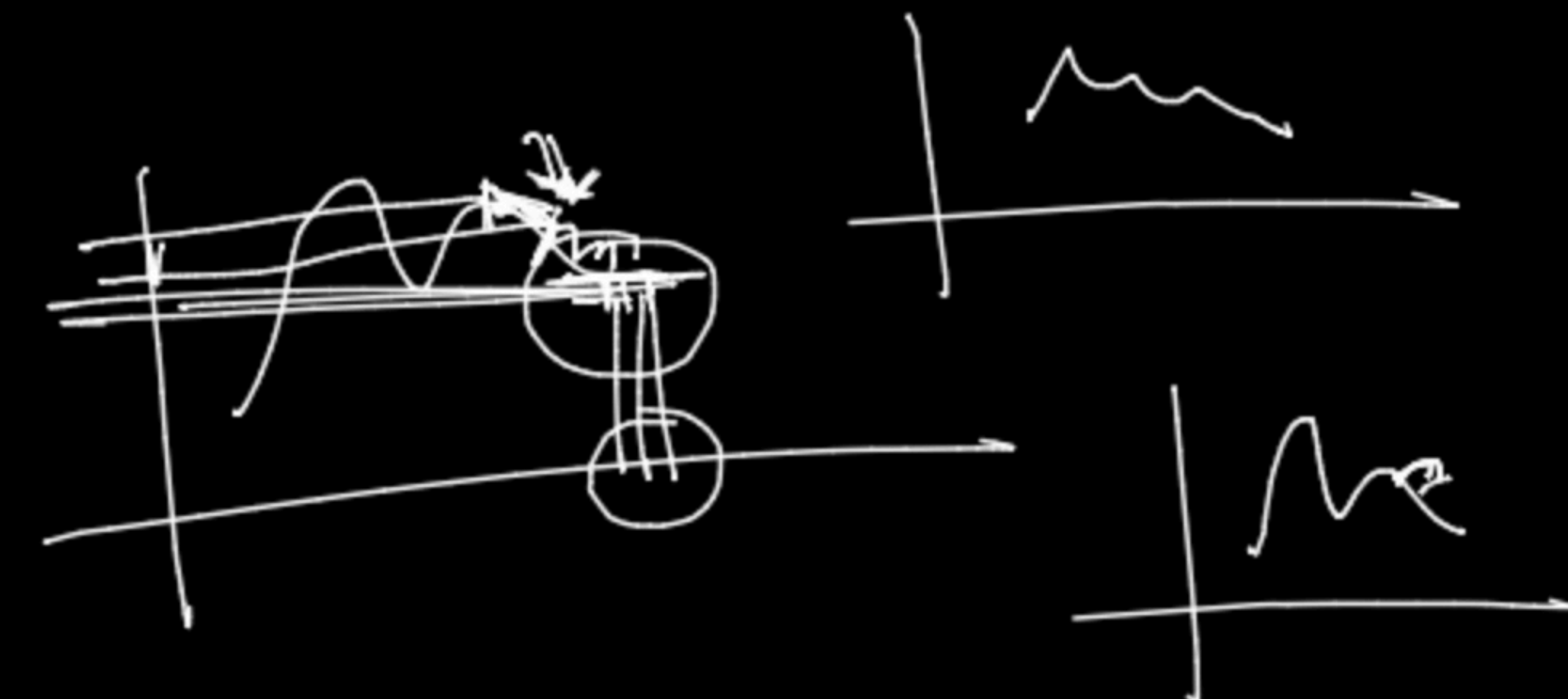
Step2 Optimisation → Algorithm

$$\min_{w_i} \sum_{i=1}^{n} (y - \hat{y})^2$$

Step3 Solve the optimisation problem with SGD / Adam

$Y = f(X)$

(3.1) initialization of w_i's → random initialization



(3.2) Compute Gradients (derivative)

$$L = (Y - w^T \cdot x_i)^2$$

$$\left(\frac{\partial L}{\partial w_1}\right) = 0 \quad \left(\frac{\partial L}{\partial w_2}\right) = 0 \quad \left(\frac{\partial L}{\partial w_3}\right) = 0 \quad \frac{\partial L}{\partial w_x} = 0$$

4 / 6

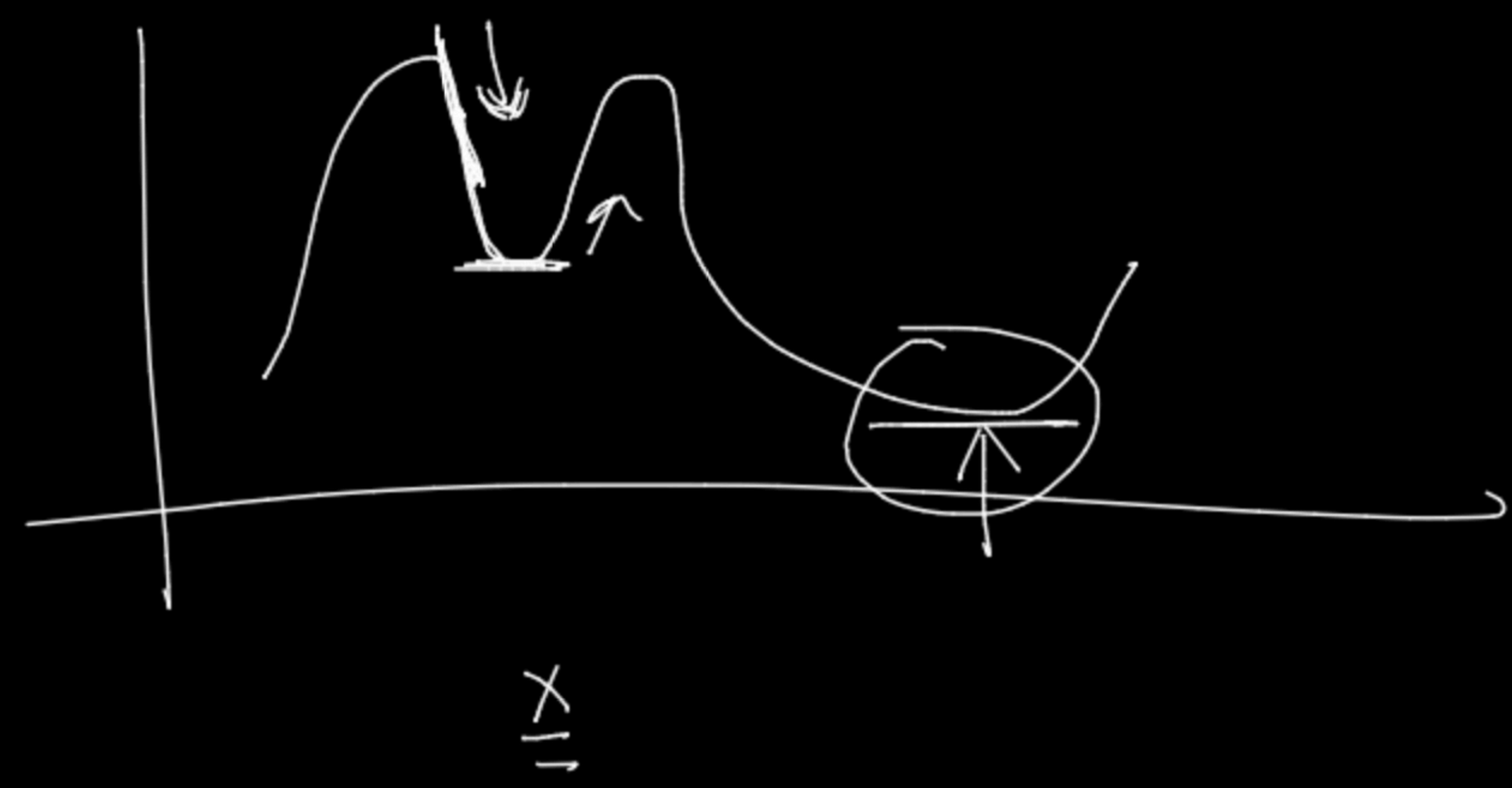
(3.3)

$$(w_{new}) = w_{old} - \eta \left[\nabla_w L \right]_{(w_{old})}$$

$$(w_{new}) \approx (w_{old})$$

$$(w_1)_{new} = (w_1)_{old} - \eta \left[\frac{\partial L}{\partial w_1} \right]$$

$$(w_1)_{new} = (w_1)_{old} - \eta \left[\frac{\partial L}{\partial w_1} \right]_{(w_1)_{old}}$$



6 / 6

x^5

$\frac{d}{dx}(x^5)$

$\frac{d}{dx} x^2$

$\frac{du}{u} \Rightarrow 0$

$5x^4$

$\frac{d}{dx} \log(x) = \frac{1}{x} \cancel{\frac{dx}{dx}}$

$\frac{d}{dx} \log(x^5) \Rightarrow \frac{1}{x^5} \frac{d}{dx}(x^5) = \frac{5\cancel{x^4}}{\cancel{x^5}} \Rightarrow \frac{5}{x}$

```
Class Student
{
    def f1()
    {

    }

    def f2()
    {
        =
        =
    }
```

use

object

```
s = Student()
s.f1()
```

8 / 8

You are screen sharing

Stop share

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Wed Jan 28 3:45 PM

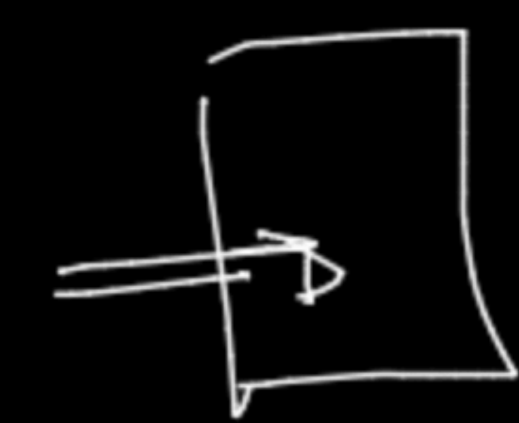
Class ATM

{ATM card-num = ______

}

def ch

def cw



atm = ATM()

atm